



भारतीय सूचना प्रौद्योगिकी संस्थान धारवाड़

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY DHARWAD

[Institute of National Importance by an Act of Parliament]

COURSE PLAN –

Session: January – April 2025 Session	
Course: Code and Title	Credits: 04
DS 358 Deep Speech Processing	
Course Instructor:	
1. Prof. S.R. Mahadeva Prasanna 2. Co-Instructors: a. Dr. Nagaraj Adiga (Associate Director, Krutrim, Bengaluru); b. Dr. Sishir Kalita (Principal Data Scientist, Armsotech, Chennai); c. Dr. Mrinmoy Bhattacharjee (Asst. Professor, Dept of CSE, IIT Jammu); d. Dr. Jagabandhu Mishra (Postdoc, Univ. of Eastern Finland)	
Teaching Assistants:	
1. Rishit Sadashiv (IIT Dharwad) and Jyoti Gadad (IIIT Dharwad)	

Course Instructor Contact Information

Office: Director's Office

Email: prasanna@iiitdwd.ac.in

Office Hours: Contact Hours apart from regular classes

1. S R M Prasanna, Any working day from 9 to 10 am (prior appointment from office)
2. Jyoti Gadad, PhD Scholar- Every Wednesday from 8am to 9am
3. Shruti B S- MTech by Research- Every Thursday from 8am to 9am

Course Schedule:

1. Deep Speech Processing – Monday and Tuesday evening 6.30pm to 8.00pm
2. Lab Evaluation Slot for Deep Speech Processing – Thursday evening 6.30pm to 8.00pm
3. Course Project Review on Monthly basis

Course Overview (or Course Objectives or Learning Outcomes)

This course aims at providing topics related to speech processing. The course contents include introduction to review of key speech processing fundamentals, machine learning methods for speech processing and finally deep learning based approaches for speech processing. The objective will be to give exposure to the latest trends in speech processing area using deep learning.

Useful Prior Knowledge

Exposure to basic concepts in linear algebra and probability

Course Syllabus

1. Overview of Speech Processing: Speech production, perception and comprehension, spoken language processing, human speech processing tasks, nature of speech signal, signal processing of speech, linguistics and phonetics, feature extraction. (6 hrs)
2. Machine Learning of Speech Processing: clustering of speech features, classification tasks in speech,

Gaussian mixture model (GMM), Hidden Markov Model (HMM), Support Vector Machine (SVM) and neural Networks (NN). (6 hrs)

3. Deep Learning of Speech Processing: Self-Supervised Learning for Speech Representation. Automatic Speech Recognition (ASR). Text to Speech (TTS) synthesis. Speaker and Language Recognition (30 hrs)

Textbooks

1. L.R. Rabiner and R.W. Schafer, Digital Processing of Speech Signals Pearson Education, Delhi, India, 2004
2. L. R. Rabiner, B. H. Juang and B. Yegnanarayana, "Fundamentals of speech recognition", Pearson Education, 2009.
3. Linus Ericsson, et al., Self-Supervised Representation Learning: Introduction, Advances and Challenges, arXiv:2110.09327
4. R. Prabhavalkar, T. Hori, T. N. Sainath, R. Schlüter and S. Watanabe, "End-to-End Speech Recognition: A Survey," in IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol. 32, pp. 325-351, 2024, doi: 10.1109/TASLP.2023.3328283.
5. Xie, Tianxin, et al. "Towards Controllable Speech Synthesis in the Era of Large Language Models: A Survey." arXiv preprint arXiv:2412.06602 (2024).
6. Wang, S., Chen, Z., Lee, K. A., Qian, Y., & Li, H. (2024). Overview of speaker modeling and its applications: From the lens of deep speaker representation learning. IEEE/ACM Transactions on Audio, Speech, and Language Processing.

Course Assessment Plan (or Course Evaluation Plan)

50% by written exams + 50% by other components

- a. Quiz-I (5%) : Four weeks
- b. Midsem (15%) : Eight weeks
- c. Quiz-II (5%): Twelve weeks
- d. Endsem (25%): Sixteen weeks
- e. Lab Assignment (15%): Continuous eval
- f. Course Project (15%): Continuous eval
- g. Surprise evaluations (10%): Continuous eval (final score weighted by percentage of attendance)
- h. Endsem viva-voce (10%)

Course Attendance Policy

Attendance Policy as per institute norms

Additional Learning Resources

- a. Speech Recognition — Deep Speech, CTC, Listen, Attend, and Spell: <https://jonathan-hui.medium.com/speech-recognition-deep-speech-ctc-listen-attend-and-spell-d05e940e9ed1>
- b. CS753, IITB, Instructor: Dr. Preethi Jyothi. Course website: <https://www.cse.iitb.ac.in/pjyothi/cs753/index.html>
- c. Gales, M., & Young, S. (2008). The application of hidden Markov models in speech recognition. Foundations and Trends® in Signal Processing, 1(3), 195-304.
- d. Alex Graves, Navdeep Jaitly, "Towards End-to-End Speech Recognition with Recurrent Neural Networks", Proceedings of the 31st International Conference on Machine Learning, PMLR 32(2):1764-1772, 2014. <http://proceedings.mlr.press/v32/graves14.pdf>
- e. Malik, M., Malik, M. K., Mehmood, K., & Makhdoom, I. (2021). Automatic speech recognition: a survey. Multimedia Tools and Applications, 80, 9411-9457.
- f. R. Prabhavalkar, T. Hori, T. N. Sainath, R. Schlüter and S. Watanabe, "End-to-End Speech Recognition: A Survey," in IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol. 32, pp. 325-351, 2024, doi: 10.1109/TASLP.2023.3328283.
- g. Speech Recognition — Deep Speech, CTC, Listen, Attend, and Spell: <https://jonathan-hui.medium.com/speech-recognition-deep-speech-ctc-listen-attend-and-spell-d05e940e9ed1>